CSET485 — ASSIGNMENT 1

STATISTICAL ANALYSIS, MODEL EVALUATION AND ALGORITHMIC FAIRNESS

Dataset : COMPAS Recidivism Dataset

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Course Code : CSET485

Submission Date : 19 October 2026

1. Introduction

In this assignment, I worked with the COMPAS recidivism dataset to understand how statistical models make predictions and how fair those predictions really are. The main goal was to see how different models behave, how the results change when we look at different demographic groups, and how the assumptions we make can affect the conclusions we draw.

First, I compared the classifications from linear regression and logistic regression. Then I ran the regression separately for different subgroups to see whether the results stayed the same. I also looked at omitted-variable bias, which shows how leaving out an important variable can change the results. After that, I tested a selected set of 50 adversarial cases to see where the model struggles. In the last part, I checked fairness using three measures: demographic parity, equalized odds, and predictive parity, at different decision thresholds.

One thing I learned is that accuracy alone is not enough to judge a model. Things like how the data is preprocessed, which threshold is chosen, how the model performs for each subgroup, and what happens when it predicts wrongly all matter a lot. Since this is observational data, I have also tried to keep in mind the uncertainty in the results and the possibility of confounding.

Overall, this assignment helped me understand the strengths and limitations of these models, and it made me think about how much evidence we would need before using recidivism predictions to make serious real-world decisions.

2. Dataset and Methodology

2.1 Dataset Overview

For this assignment, I used the COMPAS recidivism dataset to study how statistical models predict whether a person will reoffend within two years. The dataset has demographic details, criminal history variables, and the final recidivism outcome. Using these, I could check how things like prior records, age, and demographic groups are related to the predicted outcome.

2.2 Model Comparison

In the first part, I compared Linear Regression and Logistic Regression. I took 2,165 test cases and classified them using the thresholds given in the notebook. Then I checked how often the two models agreed and where they gave different results.

2.3 Subgroup Regression

Here I fitted logistic regression on the full dataset and then separately on the African-American and Caucasian subgroups. I used prior criminal-record count and age as the predictors for two-year recidivism. To understand the results, I looked at the coefficients, confidence intervals, and p-values.

2.4 Omitted-Variable Bias

In this experiment, I removed some predictors from the regression models to see how the estimated relationships changed. The variables I worked with were age, juvenile felony count, juvenile misdemeanor count, and juvenile other-offence count.

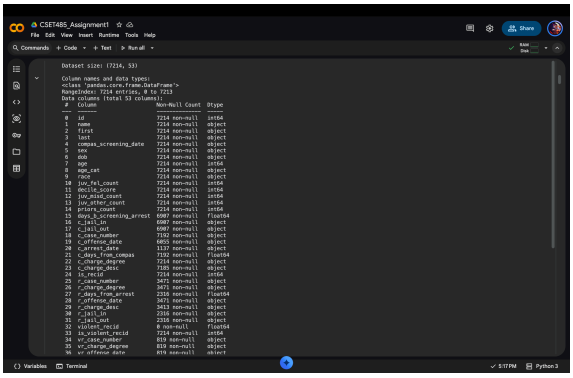
2.5 Adversarial Subset Analysis

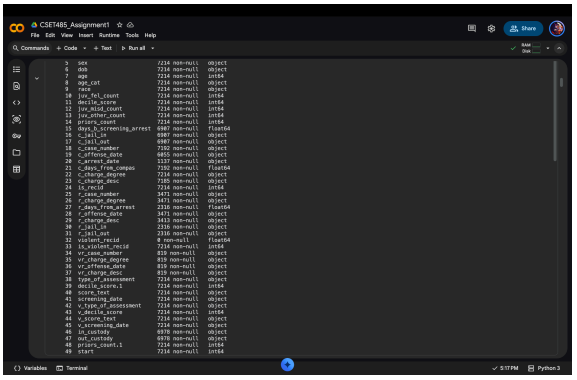
For this part, I picked a subset of 50 cases and studied its characteristics, mainly the prior criminal-record counts. Since the sample is small and was specially selected, I was careful while interpreting it because it may not represent the full dataset.

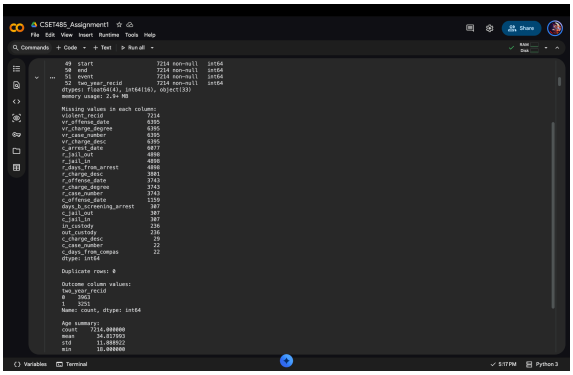
2.6 Fairness Evaluation

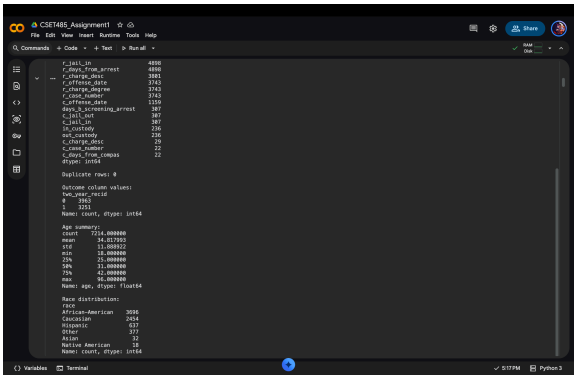
In the last analysis, I compared demographic parity, equalized odds, and predictive parity at decision thresholds from 0.05 to 0.95. The aim was to see how changing the threshold affects the fairness measures and what trade-offs come up when a model is actually used.

Screenshots :-









3. Results and Discussion

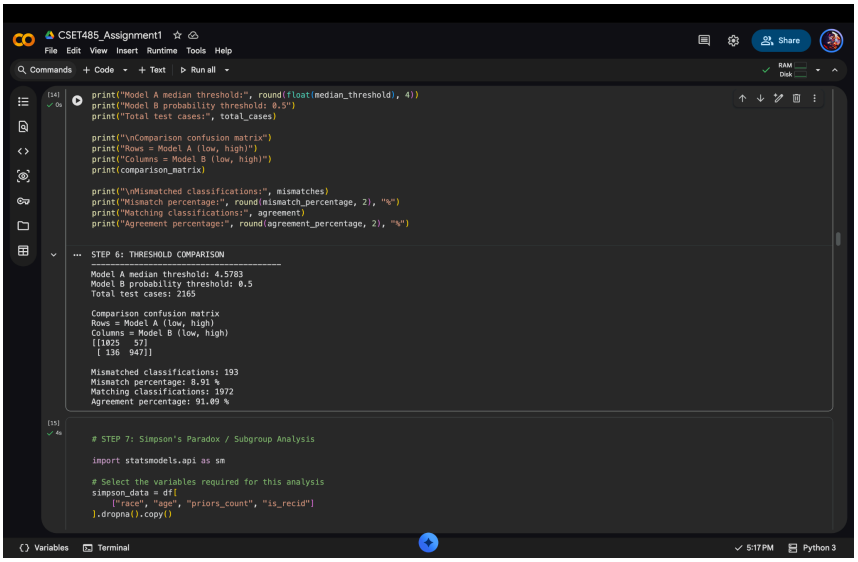
3.1 Comparison of the Two Models

I compared Linear Regression and Logistic Regression on 2,165 test cases. Both models gave the same classification for 1,972 cases, which is 91.09% agreement. The remaining 193 cases (8.91%) were classified differently.

From the comparison matrix, 57 cases were low according to Model A but high according to Model B. The other 136 cases were high according to Model A but low according to Model B. This shows that the choice of model and threshold can change the prediction for an individual person.

Linear Regression predicts a numerical value, while Logistic Regression gives a probability for a yes/no outcome. So their scales and thresholds don't mean the same thing. I think these mismatched cases should be studied properly before anyone uses such predictions for important decisions.

Screenshot :-



Model Comparison Output and Confusion Matrix

The output compares the classifications produced by Linear Regression and Logistic Regression on 2,165 test cases. The models agreed on 1,972 cases (91.09%) and disagreed on 193 cases (8.91%). The comparison confusion matrix shows the distribution of matching and mismatching classifications.

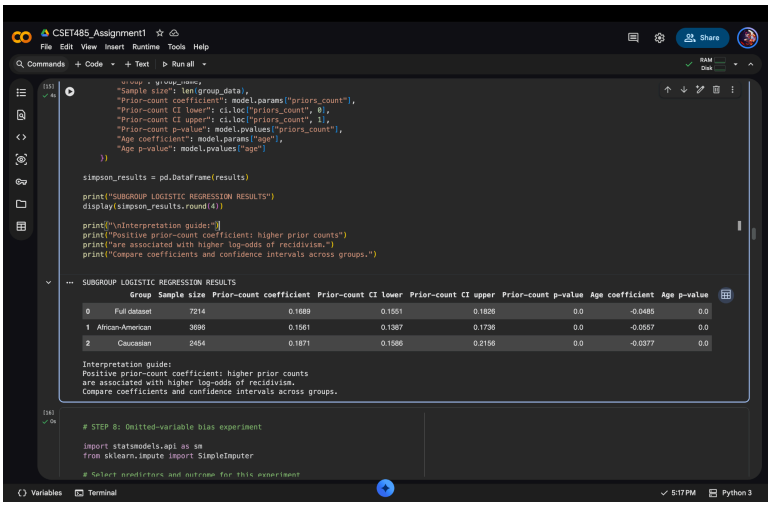
3.2 Subgroup Regression Results

The full dataset had 7,214 observations. The African-American subgroup had 3,696 observations and the Caucasian subgroup had 2,454.

The prior-count coefficient was 0.1689 for the full dataset (95% CI: [0.1551, 0.1826]). For the African-American subgroup it was 0.1561 (CI: [0.1387, 0.1736]), and for the Caucasian subgroup it was 0.1871 (CI: [0.1586, 0.2156]).

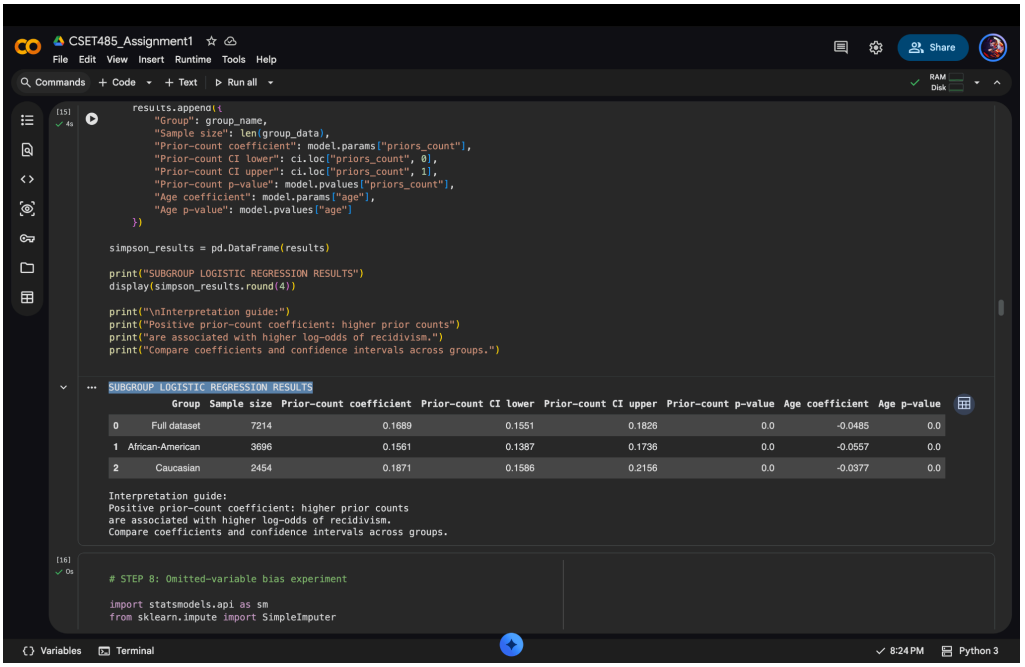
All three coefficients are positive and none of the confidence intervals include zero. This means that in these models, a higher number of prior records is associated with higher log-odds of recidivism. But this only shows association, not causation, and it does not prove that the difference between the two subgroups is statistically significant.

Screenshot :-



Subgroup Logistic Regression Results

The table compares logistic regression results for the full dataset, African-American subgroup, and Caucasian subgroup. The prior-count coefficients were 0.1689, 0.1561, and 0.1871, respectively. All three coefficients were positive, and their reported 95% confidence intervals excluded zero, indicating positive associations between prior-count values and the log-odds of recidivism in these fitted models. These results do not establish causation.



Subgroup Logistic Regression Results with 95% Confidence Intervals

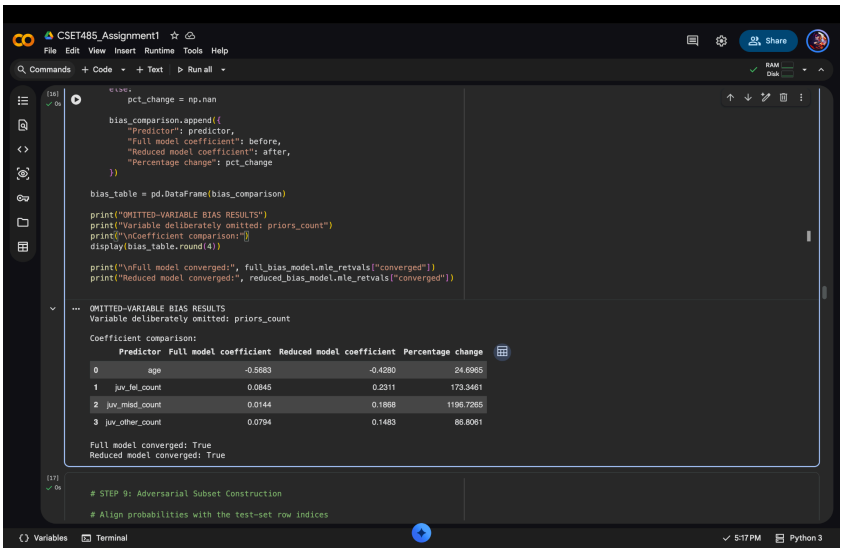
The table presents logistic regression results for the full dataset and the African-American and Caucasian subgroups. The prior-count coefficients are positive, and the reported 95% confidence intervals exclude zero. This indicates a positive association between prior-record count and the log-odds of recidivism in each fitted model. The results show association, not causation.

3.3 Omitted-Variable Bias

In this experiment, I checked how the regression estimates changed when some predictors were left out. The four predictors were age, juvenile felony count, juvenile misdemeanor count, and juvenile other-offence count.

The biggest change was 1196.7265%, for juvenile misdemeanor count. This means its coefficient was very sensitive to which variables were in the model. However, a big percentage change does not automatically prove bias, especially when the original coefficient is very close to zero. To judge it properly, we should also look at the sign, confidence interval, and significance of the coefficients.

Screenshot :-



Omitted-Variable Bias Experiment Results

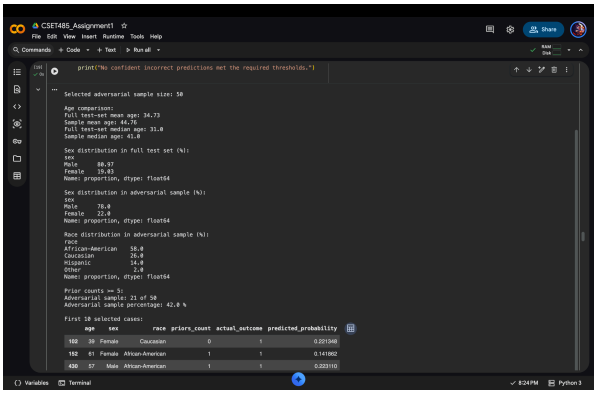
The table compares regression coefficients from the full model and the reduced model after deliberately omitting `priors_count`. The largest reported percentage change was 1196.7265% for `juv_misd_count`. Both models converged successfully. These changes demonstrate that coefficient estimates can be sensitive to the predictors included in a regression model; percentage changes should be interpreted carefully when the original coefficient is close to zero.

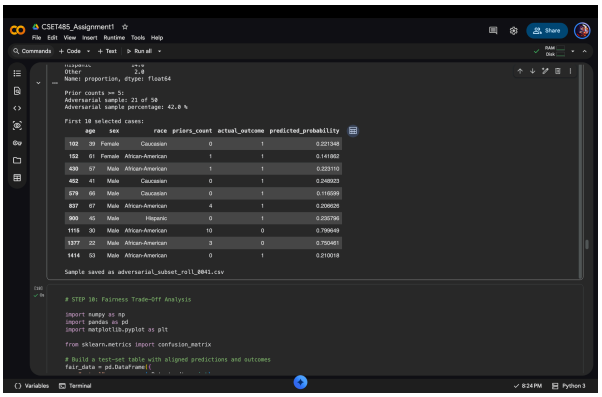
3.4 Adversarial Subset Analysis

This part looked at the 50 selected cases. Out of these, 21 cases (42%) had five or more prior records, and 29 cases (58%) had fewer than five.

So a good number of these cases had multiple prior offences. Still, having a high prior count does not mean the person will definitely reoffend. Since the subset is small and was specially selected, I can’t assume the same pattern holds for the whole population.

Screenshots :-





First screenshot — subset summary.

Second screenshot — first 10 selected cases.

Adversarial Subset Analysis Results

The analysis examined a selected subset of 50 cases. Of these, 21 cases (42%) had five or more prior records, while 29 cases (58%) had fewer than five. Because the subset was specially selected and relatively small, these findings should not be assumed to represent the entire dataset.

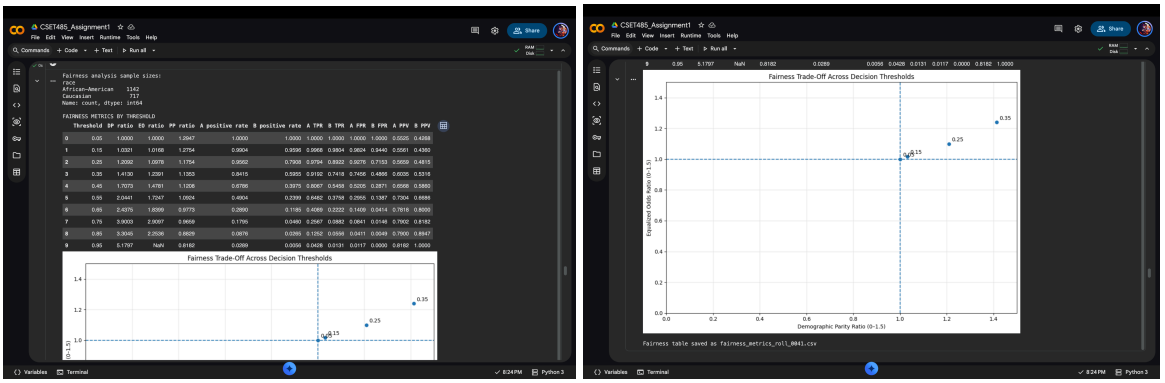
3.5 Fairness Evaluation

For fairness, I looked at demographic parity (DP), equalized odds (EO), and predictive parity (PP) at different thresholds.

At a threshold of 0.05, the DP ratio was 1.0000, EO ratio was 1.0000, and PP ratio was 1.2947. At 0.15, the values were 1.0321, 1.0168, and 1.2754. At 0.25, they were 1.2092, 1.0978, and 1.1754.

These results show that the fairness measures change depending on the threshold we choose. So no single ratio should be used alone to decide if a model is fair. We also need to consider the model's accuracy, error rates for each subgroup, uncertainty, and what happens when the model predicts wrongly.

Screenshot :-



Screenshot 1 — Fairness metrics table

It shows thresholds from 0.05 to 0.95 and the DP, EO, and PP ratios, along with group-level rates. This is the main evidence for your analysis.The table presents demographic parity, equalized odds, and predictive parity ratios at different decision thresholds, along with additional group-level performance rates.

Screenshot 2 — Fairness trade-off chart

It shows demographic parity ratios against equalized odds ratios at different thresholds. It supports your discussion of fairness trade-offs.The scatter plot illustrates the relationship between demographic parity and equalized odds ratios at selected thresholds. The dashed reference lines indicate a ratio of 1.0.

4. Conclusion and Recommendations

4.1 Conclusion

In this assignment, I worked with the COMPAS recidivism dataset and looked at model comparison, subgroup regression, omitted-variable bias, adversarial cases, and fairness measures. The main thing I understood is that judging a model is not just about checking how often it agrees with another model or how accurate it is.

When I compared Linear Regression and Logistic Regression, they agreed on 91.09% of the 2,165 test cases, but 193 cases got different classifications. In the subgroup regression, the prior-count coefficient was positive for the full dataset and for both subgroups. But this only shows association, not causation, and it does not prove that the difference between the subgroups is statistically significant.

The omitted-variable experiment showed that coefficient estimates can change a lot depending on which variables are in the model. The adversarial subset analysis showed that individual cases need to be looked at carefully, and that a small selected sample cannot be assumed to represent the whole dataset.

In the fairness analysis, demographic parity, equalized odds, and predictive parity all changed when the decision threshold changed. So no single fairness measure is enough to say that a model is fair.

4.2 Recommendations

Before using a recidivism prediction model in real life, I would recommend the following:

1. Check the model's performance on proper test data using suitable error metrics.
2. Look at subgroup performance and confidence intervals before drawing conclusions about demographic differences.
3. Test whether the coefficients stay stable when the choice of predictors is changed in reasonable ways.
4. Clearly document how the adversarial cases were selected and avoid generalizing beyond that sample.
5. Compare the fairness measures at multiple thresholds and think about what they mean in practice.
6. Keep human oversight, clear documentation, and regular monitoring after the model is deployed.

Finally, model predictions should never be the only basis for decisions that affect a person's freedom. Decision-makers also need to think about the limitations of the data, the uncertainty in the results, legal requirements, and the harm caused by wrong predictions.

5. Reflection on Model Evaluation, Fairness, and Deployment

This assignment helped me understand that checking a model's accuracy is not enough to say it is a good model. I used the COMPAS recidivism dataset to compare model classifications, look at regression results for different demographic groups, test omitted-variable bias, study a selected set of cases, and check fairness measures at different thresholds.

When I compared Linear Regression and Logistic Regression, I saw that two models can give different classifications for the same people. They agreed on 1,972 out of 2,165 test cases, but 193 cases were classified differently. This taught me that the choice of model, the prediction scale, and the threshold can all change the results.

In the subgroup regression, prior record count had a positive association with the log-odds of recidivism, but this does not mean it causes recidivism. I also learned that coefficients can change when other important variables are removed, so choosing the right variables is a big part of statistical analysis.

The fairness part felt the most important to me because predictions in criminal justice can affect a person's freedom. Demographic parity, equalized odds, and predictive parity each measure fairness in a different way. A threshold that looks good for one measure may not look good for another, so we can't decide fairness from a single ratio.

If I had to deploy a model like this, I would first test it on proper test data, check error rates for each subgroup, and make sure the data quality is good. I would also look at what happens with false positives and false negatives, write down the limitations of the model, and make sure the people using it understand how uncertain its predictions are.

I would not recommend using a recidivism prediction model as the only basis for decisions about someone's freedom. Human oversight, transparency, accountability, and regular monitoring are necessary. The model should also be checked in the setting where it will actually be used, keeping legal requirements and the harm from wrong predictions in mind.

Overall, this assignment taught me to read statistical results carefully, question assumptions, and think about the real-world effects of algorithmic decisions. A model should not be trusted as reliable or fair just because its numbers look good. It needs to be evaluated continuously.